Agenda.

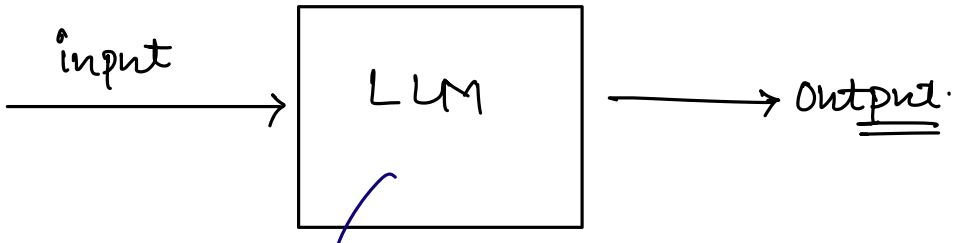
→ LLM Comparison

→ LLM cost, tokens, context window, efficiency, caching, Batching, ...

LLM | Generative AI | AI Agents | Agentic AI

LLM → Large Language Model.

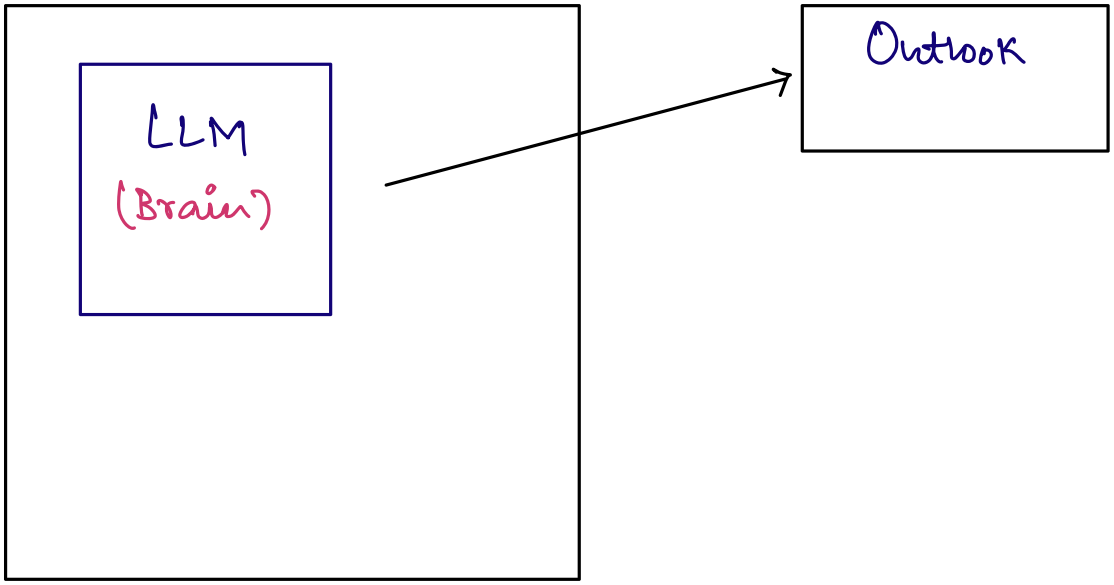
Generative AI → Generating data using LLM.




LLM → Text
Images
Videos
Audios

# AI Agents.

↳ System that doesn't only generate the answer but can also take actions.




Ex: Find the cheapest flight from DEL to BLR for 15th Mar.

→ LLM gives us the detail of flight.

→ AI Agent can

→ Book a flight

→ Recommend places | hotels.

→ ___

# Agentic AI

→ Multiple agents working autonomously

→ independently

→ Without human intervention.

Ex: Plan a 2 day trip to Jaipur in 20,000 ₹

→ LLM will just recommend the itenary.

→ Agentic AI appⁿ can

→ Compare multiple options

→ Choose one of them based on the user input

→ Book hotels

# Choosing the right LLM

→ Intelligence
- → Text generation
- → Code writing
- → Complex problem solving.

→ Cost factor
- → Tokens per dollar.

→ Latency : Time taken by the LLM to generate the response.
- → Important factor to consider for Voice | Chat agents

# LLM Cost Optimization Strategies.

1) Token Counting

↳ How much text our LLM is processing.
(input + output)

Token Count ↑ ⇒ Cost ↑

2) Caching

↳ Cache (Store nearby) answers for some of the most frequent queries so that we can answer these queries from the Cache itself without making a call to LLM.

3) Prompt Optimization

4) Batch Processing

↳ Non real time tasks can be processed in batches in order to optimise cost.

→ Overnight Report generation

→ Bulk Content creation

→ Chat | Voice agents ✗

5) Two-tier Processing